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Carbon condensation wave in C_3O_2 and C_2H_2 initiated by a shock wave

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Abstract

Formation of the waves of condensation of carbon nanoparticles at self-decomposition C_3O_2 and C_2H_2, initiated by a shock wave, was experimentally observed. It was found that self-sustaining condensation wave in C_2H_2 could be formed at much higher pressures than in C_3O_2 despite essential larger heat release of the process $C_2H_2 \rightarrow H_2 +$ *soot* comparably to $C_3O_2 \rightarrow 2CO +$ *carbon nanoparticles*. The basic kinetic characteristics of process of pyrolysis and following growth of nanoparticles – induction time and rate constant of particle formation are determined for both mixtures. A connection between various stages of process and heat release of condensation is analyzed. It is shown that induction time in both mixtures is not accompanied by a noticeable heat release. For C_2H_2, duration of this stage is approximately two orders of magnitude larger than for C_3O_2. Against that, the following stage of carbon nanoparticle formation, accompanied by an essential heat release, proceeds in C_3O_2 and C_2H_2 with comparable rates. It is demonstrated that distinction in formation of a self-sustained condensation wave in C_3O_2 and C_2H_2 is caused by the long induction time in acetylene reflecting the growth of large polyhydrocarbon molecules. The increase in pressure leads to narrowing of this zone and a collapse of whole process to a detonation wave of condensation.
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1. Introduction

The processes of pyrolysis of most gaseous hydrocarbon fuels are accompanied by the formation of condensed carbon particles and the process of condensation, like the oxidation reactions, is characterized by the essential heat release. Therefore, the challenge is to realize conditions where the process of nanoparticle formation yielded during pyrolysis forms a self-sustaining wave of condensation, propagating similarly to detonation wave.

A well known process of this kind is the wave of acetylene self-decomposition propagating either by combustion [1–3] or by detonation [4–6] wave. The main peculiarity of acetylene decomposition wave yielding the condensed carbon particles is underlined in all these papers. These works show an extremely important role of the energy of carbon condensation for the formation of self-decomposition waves. The integral heat release of the process $C_2H_2 \rightarrow H_2 +$ *solid graphite* is 234 kJ/mol, which is

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close to the heat release of hydrogen oxidation reaction (242 kJ/mol). However, an extraction of quantitative input of condensation energy for forming the self-sustaining detonation wave in acetylene is hampered by the complex intermediate reactions of various polyatomic hydrocarbons formation during acetylene decomposition [7,8].

From this point of view, the pyrolysis of the hydrogen-free substance carbon suboxide C_3O_2 is much plainer way to study the heat release of carbon condensation and the possible influence of this energy on the propagation of a shock wave, initiating the pyrolysis process. Carbon suboxide is a rather unstable volatile compound decomposing to carbon atom and two CO molecules at 1400–1600 K. Therefore, the process of C_3O_2 decomposition and subsequent carbon condensation proceeds at the total absence of secondary gaseous reactions (in the system, only CO remains, chemically stable at T <4000 K). The effective rates of carbon particle formation, observed in C_3O_2 pyrolysis [9–13], are substantially higher than those obtained for all hydrocarbons [14–18].

In our previous paper [13], the considerable heating of the mixtures initially containing only 3% C_3O_2 in Ar was observed – the complete transformation $C_3O_2 \rightarrow 2\ CO + carbon\ nanoparticles$ behind the shock wave resulted in the additional temperature rise up to 300 K. Based on these results we later performed experiments in richer mixtures containing 10–30% C_3O_2 in Ar. In this case, the essential acceleration of the shock wave and formation of detonation-like condensation wave was observed [19]. Therefore, it is rather interesting to investigate the possibility of formation of a similar detonation wave of condensation initiated by a shock wave in resembling C_2H_2/Ar mixtures.

Thus, the goal of this work was the experimental study of formation of carbon condensation waves, initiated by a shock wave in C_3O_2 and C_2H_2 and analysis of the distinctions and similarities in kinetics of carbon particle formations and their final properties in both these mixtures.

2. Experiments

Experiments were performed behind reflected shock waves in a diaphragm-type shock tube with an inner diameter of 70 mm, a 3.5-m long driver section and a 4.5-m long driven section. The propagation of reflected shock wave in the mixtures initially containing various amounts of C_3O_2, H_2 and C_2H_2 in Ar was studied. The experimental conditions for all series are summarized in Table 1.

The actual pressure and shock wave velocity were measured by several piezo-gauges installed at the various distances 0–300 mm from the end plate of shock tube. For the observation of shock wave behavior, the shock tube was equipped by two rectangular sapphire windows 160 mm × 5 mm installed on the distance 25 mm from the end plate. Through these windows, the time resolved images of radiation behind shock wave in the range 300–800 nm were recorded using an ICCD camera (StreakStar II, LaVision GmbH). The laser extinction at $\lambda = 633$ nm, depicting the condensed particle formation, was registered through the same windows at the different distances from the end plate. The test experiments were performed in the mixture 1% C_6H_6 + Ar at the temperature $T_5 = 2800$ K. In these conditions, pyrolysis goes instantly, particles grow quickly (during ~10 μs), and heat effect is insignificant. These records proved the wave moving with the constant velocity, pressure gauges showed good plateaus and all measured values well coincided with calculations.

In addition to these on-line measurements, the samples of particles formed during experiments, were studied by an electron microscope CM 30, Phillips. For studying the size distribution of the particles, a low magnification of $m = 45{,}000$ was chosen. For obtaining the morphology and the structure of the particles, they were observed at high resolution where a magnification of $m = 400{,}000$ and 490,000 was applied. To determine the degree of crystallization of the particles, micro diffraction measurements (MDF) were partly performed. The elementary analysis for

Table 1
Investigated mixtures, ranges of initial pressure P_5 and temperature T_5 behind reflected shock waves and symbols used in the figures.

Series	Mixture	P_5 (bar)	T_5 (K)	Symbol
A	10% C_3O_2 + Ar	4.1–4.5	1540–1680	⬟
B	20% C_3O_2 + Ar	7.5–7.7	1470–1550	▼
C	30% C_3O_2 + Ar	8.0–8.2	1425–1490	♦
D	20% C_3O_2 + 10% H_2 + Ar	6.9–7.2	1490–1510	×
E	31% C_3O_2 + 6% H_2 + Ar	7.1–7.6	1505–1535	✳
F	20% C_2H_2 + Ar	5.5–6.4	1440–1970	●
G	20% C_2H_2 + Ar	7.9–14	1505–2960	■
H	20% C_2H_2 + Ar	29.9–30.6	1310–1330	▲
I	30% C_2H_2 + Ar	5.6–7.1	1307–1780	+

obtaining the H/C-ratio in the particle content was performed with a CHNS-El3 apparatus (Elementar Vario).

3. Results

3.1. Formation of condensation wave

To study the peculiarity of formation of carbon condensation wave at hydrogen-free conditions and for substances containing hydrogen, three different mixtures were investigated. For hydrogen-free conditions, a mixture containing 20% C_3O_2 in Ar was selected (series B), where the formation of Chapman–Jouguet detonation wave of condensation was already observed [19]. To study the influence of free hydrogen on the carbon condensation, a mixture of 20% $C_3O_2 + 10\%\ H_2$ in Ar (series D) was chosen. Since each C_3O_2 molecule yields one C atom, the condensation process in such a mixture proceeds at equimolecular C/H conditions. To analyze the formation of condensation wave in the bound carbon–hydrogen species, the mixtures of different amount of C_2H_2 in Ar (series F–I) were studied.

In Fig. 1, the records of ICCD camera for all these mixtures are shown. Figure 1a–c exhibit the behavior of the shock wave at similar initial parameters behind reflected shock wave: $T_5 = 1600 \pm 100$ K and $P_5 = 6.5 \pm 1$ bar (series B, D and F). In the first two mixtures, containing C_3O_2 or $C_3O_2 + H_2$ (B and D), the development of the detonation-like condensation wave proceeds in a very similar way, the bright radiation zone, emerging behind the shock front, at short distance from the end plate catches it up, resulting in intensive acceleration of the shock wave accompanied by the formation of sharp peaks of radiation. In contrast, the acetylene mixture F demonstrates the steady propagation of the complex of a shock wave and radiation zone (reflecting the heat release processes of particle formation) on some constant distance behind it. In other words, in this case, the propagation of condensation wave is completely controlled by a shock wave, but the inverse influence of the condensation on the shock wave parameters does not take place.

To reduce a total time of process and to obtain a detonation-like wave of condensation in the acetylene, similar to carbon suboxide, it seemed natural to increase intensity of a shock wave, i.e. initial temperature of pyrolysis. However, the experiments carried out at $T_5 = 1900 \pm 100$ K and the same pressure did not reveal noticeable change in behavior of a shock wave – the time delays of particle formation became somewhat shorter, but both, the shock wave and a condensation wave, still moved with constant speed on the fixed distance from each other. The following attempt consisted in increasing the C_2H_2 content of a mixture. Again, these experiments performed in the mixtures, containing 30% and even 40% C_2H_2 in Ar, did not led to essential change in character of shock wave propagation and the subsequent processes of particle growth.

However, an increase of pressure behind shock wave immediately gave a convincing picture of formation of detonation-like wave of self-decomposition of acetylene. Acceleration of a shock wave and its transition in a detonation-like mode of propagation took place at pressure $P_5 = 30$ bar even at initial temperatures $T_5 = 1320$ K (series H, see Fig. 1d). In this case, the picture of shock wave propagation is rather similar observed in the mixtures B and D (Fig. 1a and b).

Figure 2 demonstrates the typical change of pressure, radiation and extinction time-profiles, observed during formation of detonation-like condensation wave for all mixtures. In upper records (a), immediately after achievement of

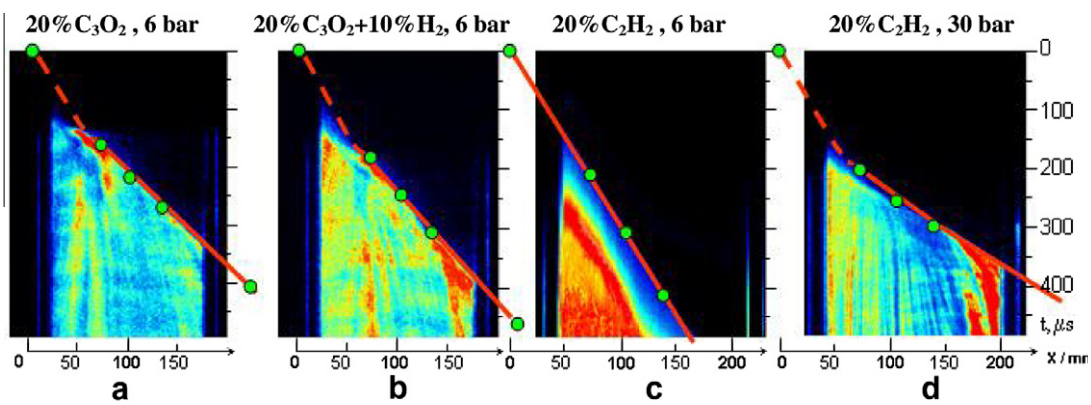

Fig. 1. ICCD images of shock wave propagation in various conditions. Points and lines show the trajectories of shock wave front measured by the pressure gauges.

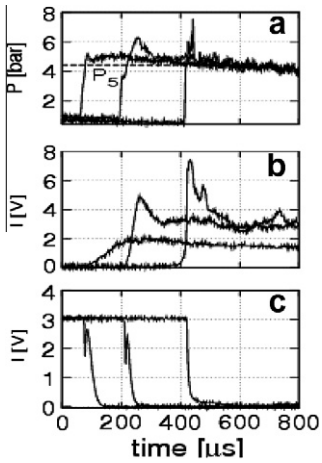

Fig. 2. Typical example of change of pressure (a), radiation (b) and extinction time-profiles (c), observed during formation of detonation-like condensation wave (series A).

calculated values of pressure behind a shock wave front (dashed line), an additional pressure rise can be observed. Further propagation of shock wave is characterized by the appearance of sharp pressure peak just behind the front. These processes are accompanied by the rise of radiation peaks (records (b)) testifying to essential temperature increase in the narrow zone behind the wave front. Lower records (c) demonstrate a sharp rise of extinction, reflecting the formation of condensed particles. (The narrow peak observed on the first two plots at the moment of shock front propagation, is caused by the schlieren effect on the density gradient.) Obviously, the condensation process is also accelerating with the propagation of the wave. Analysis of these time-profiles enabled to study the kinetics of particle formation in all investigated mixtures.

3.2. Kinetics of particle formation

The extinction measurements, performed for all series of experiments, allowed determining the main kinetic characteristics of process of formation of the condensed carbon particles – the induction time and the effective rate constant of particle formation. The induction time τ_{ind} was defined as the intersection of the inflectional tangent of the extinction profile with the time axis. The rate constant of particle formation k_f was determined by approximation of the later part of extinction profiles by the relaxation equation

$$\frac{dD}{dt} = k_f \cdot (D_{st} - D), \tag{1}$$

where D is the normalized optical density, which was calculated using Lambert–Beer law:

$$D = \frac{\ln(I/I_0)}{[C] \cdot l}, \tag{2}$$

I_0 and I_1 are the incident and transmitted light intensities, D_{st} is optical density value measured at the late, stationary stage of extinction, $[C]$ is the total available carbon concentration, and l is the shock tube diameter.

Since during formation of detonation-like regimes, the parameters behind shock wave changed essentially, for kinetic studies, only extinction profiles measured near the end plate of shock tube, where these changes did not started yet, were taken. In Fig. 3, the induction times τ_{ind} in the investigated mixtures were plotted against the "frozen" temperature T_5 compared to data from literature. Since τ_{ind} is basically the phenomenological characteristic of the process and cannot be attributed directly to any concrete chemical reaction during pyrolysis, its dependence on carbon concentration and temperature is not clear a priori. The data presented in all works published, testify for considerable divergence of τ_{ind} obtained at different carbon concentrations. Attempts to generalize the dependences led authors to very different power factors α in dependence of $\tau_{ind} \sim [C]^{-\alpha}$. A number of authors confined to the most obvious value $\alpha = 1$ [15,20,21], however, authors [14] used for C_2H_2 $\alpha = 2$, while work [22] results for C_2H_2 a much less value $\alpha = 0.186$, authors of [17] suggests for C_2H_2 $\alpha = 0.75$; for C_3O_2 $\alpha = 0.5$ is recommended in [9], etc. Summarizing a considerable quantity of different data, the most reasonable value $\alpha = 1$ is used here (see Fig. 3). All deviations from this dependence are most likely caused by the disregarding of changes of optical properties of particles at an early stage of their

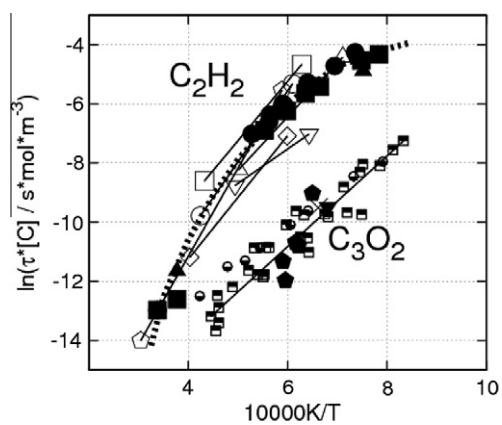

Fig. 3. Measured induction times of extinction rise (symbols see Table 1) in comparison with the data of previous works for highly diluted mixtures: ⌂ – 10% C_2H_2, [14]; □ – 5% C_2H_2, [15]; ○ – (0.2–1%) C_2H_2, [17]; △ – 2% C_2H_2, 10–12 bar, [21]; ▽ – 2% C_2H_2, 1–2 bar, [21]; ◇ – (1–20%) C_2H_2, [22]. ⬓ – (0.3–1%) C_3O_2, [13]; ◒ – (0.33% C_3O_2 + 0.33% H_2), [13]. Curve dotted line – approximation of the present data.

growth [11] which just corresponds to induction time.

The data for C_3O_2 and for $C_3O_2 + H_2$ fit well with the data of previous measurements in much more diluted mixtures. Note that the experiments performed in the mixtures 20% $C_3O_2 + 10\%$ H_2 confirmed results [9] indicating that hydrogen addition in C_3O_2 does not change the induction time. The induction times measured in the mixtures containing 10–30% C_2H_2 also showed a good agreement with numerous previous works [14,15,17,21,22]. One can see from Fig. 3 that the induction times for C_3O_2 and C_2H_2 in the temperature range 1250–2500 K differ by two orders of magnitude. However, based on all these data a non-linear temperature dependence $\ln\{\tau_{ind}[C]\} \sim T^{-1}$ could be assumed (curve dotted line in Fig. 3), which denotes a sharp acceleration of the process toward higher temperatures.

In the following Figs. 4 and 5, the rate constants of particle formation k_f in the investigated mixtures are given. Since most literature data on particle growth rates, as well as the induction times, are well generalized by the dependence $k_f \sim [C]$, they are presented in the similar coordinates: $\ln\{k_f/[C]\} \sim T^{-1}$.

Unlike the induction times, k_f measured in the given work, differ from experimental results obtained for highly diluted mixtures. In case of C_3O_2 it shows tenfold larger value in comparison to the data for highly diluted mixtures (Fig. 4). In C_2H_2 this difference is stronger and measured k_f exceeds data for (0.2–1%) C_2H_2 more than in 30 times (Fig. 5). Note that, due to an essential increase in particle growth rates for C_2H_2 relatively to highly diluted mixtures, these rates become comparable to those for C_3O_2. Therefore,

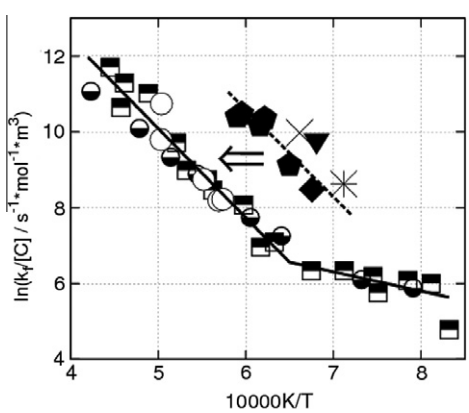

Fig. 4. Rate constants of particle formation measured in C_3O_2 (symbols see Table 1) compared with measurements in highly diluted mixtures (○ – 3%C_3O_2 [13], ■ – (0.3–1%) C_3O_2 [9], ◒ – 0.33% $C_3O_2 + 0.33\%$ H_2 [9]. Horizontal arrow shows the possible shift of the real temperature due to heat release of particle formation.

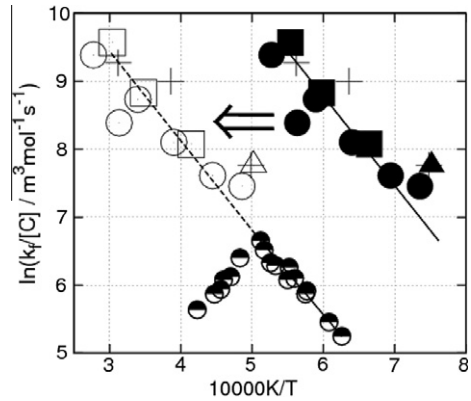

Fig. 5. Rate constants of particle formation measured in C_2H_2 (symbols see Table 1) compared with measurements in highly diluted mixtures (◓ – (0.2–1%) C_2H_2 [17]. Horizontal arrow and large empty points show the possible shift of the present data on the temperature scale due to heat release of particle formation.

it is obviously that just the longer induction time for acetylene in comparison to C_3O_2 leads to a delay of heat release and a decreasing efficiency of its transfer to energy of a shock wave.

3.3. Particle analysis

In all experiments performed, a sufficient amount of solid particulate material was formed. In Fig. 6, the typical TEM images of yielded carbon particles are shown. In the hydrogen-free mixture B, the particles are large having mean diameter $\bar{d} = 30$ nm, while the particles in the mixtures D and H are much smaller, with mean diameters $\bar{d} = 18.4$ nm and $\bar{d} = 19.3$ nm correspondingly. The high resolution TEM analysis showed completely amorphous particles from mixture B, while in mixtures D and H the increasing degree of crystallization could be observed.

The analysis of the relative content of carbon and hydrogen in these particles showed that they consist almost completely of carbon; the maximum content of hydrogen in the particles from all mixtures was less than the limit of error (~0.3%).

4. Discussion

To analyze the reasons of distinction of formation of a detonation-like wave of condensation in C_3O_2 and C_2H_2, it is necessary to consider the different kinetics of exothermic processes of growth of the condensed carbon particles in these two systems.

The most basic differences between the mechanism of carbon nanoparticle formation at pyrolysis of hydrogen-free precursor C_3O_2 and soot

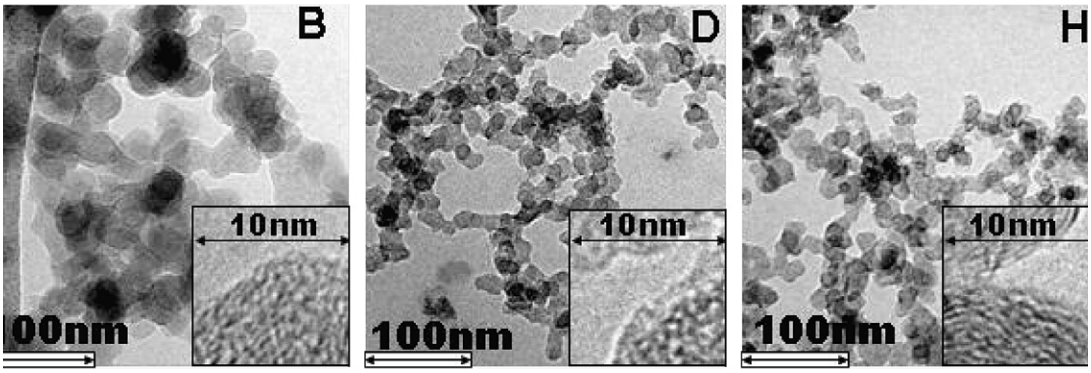

Fig. 6. Low and high resolution (insertions) transmission electron microscope images of carbon nanoparticles formed in detonation-like waves in series B, D and H.

formation at pyrolysis of any hydrocarbons is that the primary products of C_3O_2 pyrolysis are straight carbon atoms and radicals C_2 [23]:

$$C_3O_2 + M \rightarrow C_2O + CO + M \quad (3)$$

$$C_2O + M \rightarrow C + CO + M \quad (4)$$

$$C_3O_2 + C \rightarrow C_2 + 2CO \quad (5)$$

$$C_2O + C_2O \rightarrow C_2 + 2CO \quad (6)$$

$$C_2O + C \rightarrow C_2 + CO \quad (7)$$

Therefore, the further growth of carbon clusters and formation of condensed nanoparticles begin directly with pyrolysis reactions and proceeds either by surface growth

$$C_n + C \rightarrow C_{n+1}, \quad (8)$$

or by coagulation of small clusters

$$C_n + C_m \rightarrow C_{n+m}, \quad (9)$$

with rates, comparable with frequency of gas kinetics collisions [24].

In contrast to C_3O_2, acetylene pyrolysis, as well as any other hydrocarbons, does not yield directly the carbon clusters. For example, in work [22] as primary reactions of acetylene pyrolysis, the reactions of H abstraction

$$C_2H_2 + M \rightarrow C_2H + H + M \quad (10)$$

and polymerization of C_2H_2

$$C_2H_2 + C_2H_2 \rightarrow C_4H_3 + H \quad (11)$$

are discussed under various pyrolysis conditions. During the further reactions the complex hydrocarbon radicals either polyacetylenes [25,26], or polyaromatics [7,26,27] are formed.

Anyhow, as it has been shown in [11], the hydrocarbon particles formed at C_2H_2 pyrolysis, as well as young carbon particles formed in hydrogen-free C_3O_2 mixtures, are transparent for the probing laser on 633 nm and just this stage is reflected by the long period of induction in acetylene mixtures. Since the data of the given experiments plotted against the "frozen" temperature T_5, well coincides with the data for highly diluted mixtures (see Fig. 3), one can conclude that the heat release at these stages is not significant yet and the temperature of a mixture does not noticeably differ from initial T_5.

The further stage of process, characterizing by formation of carbon particles, non-transparent on 633 nanometers, has in present experiments the considerable discrepancies with the data of previous works (see Figs. 4 and 5). It is the most natural assuming the observed difference of the measured particle growth rates with the highly diluted mixtures is caused by an essential heat release at this stage of the process, resulting in a real temperature rise of a mixture. Horizontal arrows in Figs. 4 and 5 demonstrate the possible shift of the data on the temperature scale due to the change of the real temperature. Estimations show that temperature growth of a mixture at this stage of process can reach about 200 K in mixture B and about 1000 K in mixture H. The contradiction with the data [17] showing at these temperatures a fall of particle growth rates in C_2H_2, can be interpreted according to [12,28], specifying in two-stage particle formation at T above 2200 K. At low concentration of precursor, the first, accelerating stage of particle formation becomes less and less noticeable on extinction signals because of decreasing size of primary particles. Therefore, the data [17] reflects only secondary growth of particles, slowing down with the temperature rise. At the conditions of the given work, a volume fraction of particles is much higher, consequently, the obtained data characterizes a primary particles growth.

Such an interpretation of distinction of thermal effects at an extinction growth stage is qualitatively illustrated in Fig. 7 where the approximate values of integral heat balances of processes of pyrolysis C_3O_2 [19] and C_2H_2, depending on the current

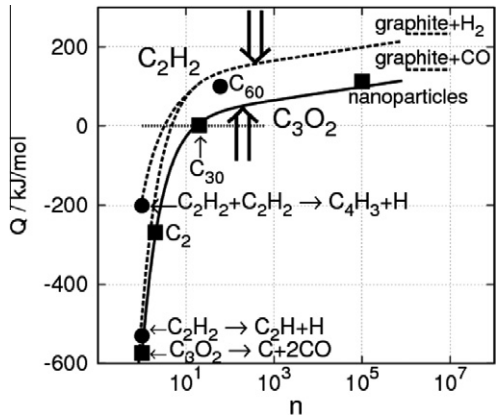

Fig. 7. Approximate plots of the heat balance during pyrolysis of C_3O_2 [19] and C_2H_2 in dependence on number of carbon atoms in the forming particles.

particle sizes, are given. Vertical arrows specify estimations of particle sizes at which an essential increase in refractive index on 633 nm takes place. In case of pure carbon particles owing to larger refractive index [9–11], extinction starts growing at the smaller sizes.

Discussion about the path of a transition from polyhydrocarbon molecules to carbon nanoparticles proceeds till now [26]. The prevailing hypothesis is based on the mechanism of HACA [27] consisting in gradual growth of PAH and decreasing of quantity of hydrogen atoms in them. However, this hypothesis can not describe a growth of carbon particles in hydrogen-free systems which concerns C_3O_2. Results of the given experiments show that the rate of formation of particles and their final properties in both investigated mixtures are close enough. This fact evidences for weakly dependence of the last stages of particle growth on precursors type. Therefore the presence of free hydrogen on the last stages likely only slows down of particle coagulation, limiting a growth of their final size.

Thus, fast development of a detonation wave of condensation in C_3O_2 is defined by that a "bottle-neck" of the whole process is endothermic reaction of decomposition of initial molecules, exponentially accelerating with temperature rise. This reaction directly gives start to exothermic processes of carbon cluster growth which in all cases occurs extremely quickly and has as though character of "explosive condensation". Therefore the regularities of formation of such a wave in C_3O_2 are in many aspects similar to usual processes of deflagration–detonation transition. On the other hand, the key stage of process of acetylene decomposition, determining the possibility of formation of a detonation wave of condensation, is the long stage of polymerization of hydrocarbons. This stage of process is not accompanied by a noticeable heat release, however, separates spatially front of a shock wave and a condensation zone, thereby interfering with an influence of heated gas layers on shock wave parameters. The increase in pressure and, accordingly, concentration of reacting particles leads to narrowing of this zone and its coupling with the zone of heat release. The following temperature rise in this zone results in sharp decrease of induction time (see dotted curve in Fig. 3) and collapse of whole process to a detonation wave of condensation.

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